Hall Ticket Number:

MR-22

Code: B325CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Computer Organization and Architecture

(Common to CSE, CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1= 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define interrupt?	1	1	1	1
1. b)	What is the use of load and store memory reference instruction?	1	1	1	6
1. c)	Discuss about the term control word?	1	1	2	1,3
1. d)	Differentiate hardwired control and micro-programmed control.	1	1	2	1
1. e)	Draw the flow chart for the add and subtract operations.	1	6	3	5,2
1. f)	Write about various decimal arithmetic micro operation symbols.	1	1	3	1
1. g)	Define burst transfer.	1	1	4	5
1. h)	Define Hit ratio and Miss ratio.	1	1	4	1
1. i)	What is Microprocessor?	1	1	5	1
1. j)	Explain the advantages of pipelining.	1	2	5	5

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Analyse the Basic Operational concepts of Computers.	5	4	1	1
2. b)	List and explain the steps involved in the execution of a complete instruction cycle. List the basic symbols used in register transfer Language.	5	4	1	3,4
	OR				
2. c)	Contrast memory reference instructions with respect to register reference instructions.	5	4	1	2
2. d)	Define bus structures. Explain about the bus structures of computers.	5	2	1	2
3. a)	Explain Address sequencing in micro programmed control unit?	5	2	2	2,3
3. b)	Illustrate the micro-programmed control unit (MCU) with a neat diagram and how instructions will be processed in MCU?	5	2	2	1
	OR				
3. c)	Explain different types of addressing modes.	5	2	2	1
3. d)	Define an instruction format? Explain different types of instruction formats in detail.	5	2	2	1,4
4. a)	With a neat Flowchart analyse Floating-Point Addition and Subtraction.	5	4	3	4,5
4. b)	Explain the hardware implementation for signed magnitude for addition and subtraction.	5	2	3	5,2
	OR				
4. c)	Elaborate steps used in Booth algorithm for multiplying following binary numbers (+15) * (-13).	5	6	3	4,5
4. d)	Demonstrate division algorithm for fixed point numbers with the help of a flow chart.	5	2	3	5,2
5. a)	What is auxiliary memory? Explain the various memory components used as auxiliary memory in computer systems.	5	4	4	1
5. b)	Explain the following with respect to asynchronous data transfer. i) Strobe control ii) Handshaking iii) Asynchronous serial transfer.	5	2	4	1
	OR				

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5. c)	Explain the Daisy Chaining priority with neat diagram.	5	2	4	1,2
5. d)	Explain DMA Controller in detail with neat diagram?	5	4	4	2,3
6. a)	What is meant by instruction pipeline? Explain four segment Instruction Pipeline. Give its timing diagram.	5	2	5	3
6. b)	What are the different major conflicts arising in pipelined execution.	5	5	5	3
	OR				
6. c)	Explain the characteristics of multiprocessors.	5	2	5	5
6. d)	In detail describe multiport memory, cross bar switch and multistage switching networks used for interconnecting multiprocessors.	5	4	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome